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The Effects of 8-Week Creatine Hydrochloride and Creatine Ethyl Ester Supplementation on Cognition, Clinical Outcomes, and Brain Creatine Levels in Perimenopausal and Menopausal Women (CONCRET-MENOPA): A Randomized Controlled Trial

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The Effects of 8-Week Creatine Hydrochloride and Creatine Ethyl Ester Supplementation on Cognition, Clinical Outcomes, and Brain Creatine Levels in Perimenopausal and Menopausal Women (CONCRET-MENOPA): A Randomized Controlled Trial

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ABSTRACT

Objective: To investigate the effects of an 8-week supplementation with varying doses of creatine hydrochloride, administered alone or in combination with creatine ethyl ester, on cognitive function, clinical outcomes, brain creatine concentrations, and biochemical markers in perimenopausal and postmenopausal women. This study specifically aimed to evaluate the potential of low-dose creatine formulations with enhanced solubility as a targeted intervention for menopause-related neurocognitive and metabolic changes.

Methods: A total of 36 apparently healthy perimenopausal and menopausal women (mean age 50.1 ± 5.7 years) were randomly allocated to one of four groups: low-dose creatine hydrochloride (750 mg/day), medium-dose creatine hydrochloride (1,500 mg/day), creatine hydrochloride plus creatine ethyl ester (800 mg/day), or placebo, in this randomized controlled double-blind trial. Menopausal women were defined as having no menstrual cycle for 12 consecutive months without other causes, while perimenopausal women were still menstruating but reported at least one symptom such as hot flashes, sleep disturbances, mood swings, or concentration difficulties.

Results: Supplementation with medium-dose creatine hydrochloride was found to be superior to placebo in enhancing reaction time (1.2 vs. 6.6%; $p < 0.01$), increasing frontal brain creatine levels (0.9 vs. 16.4%; $p < 0.01$), and favorably modulating serum lipid profiles ($p < 0.05$). Moreover, medium-dose creatine hydrochloride demonstrated a potential advantage over other treatments in reducing the severity of mood swings ($p = 0.06$). All interventions were well tolerated, with no severe adverse effects reported.

Conclusion: Our findings suggest that this supplementation protocol may be a promising, safe, effective, and practical dietary strategy for improving clinical outcomes and elevating brain creatine concentrations in perimenopausal and menopausal women. The trial was registered at ClinicalTrials.gov (NCT06660004).

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Introduction

Creatine (2-[carbamimidoyl(methyl)amino]acetic acid) is a naturally occurring, semi-essential nutrient that plays a pivotal role in buffering and regulating cellular bioenergetics throughout the human body (1). Initially recognized for its ergogenic benefits in athletic performance during the early 1990s, creatine research has since expanded into broader areas of human nutrition and diverse populations. Over the past two decades, creatine has emerged as a subject of interest in female physiology and health, spanning exercise performance, reproductive health, and clinical medicine. Hormone-driven alterations in endogenous creatine synthesis, transport, and enzymatic machinery suggest that substantial changes in this cellular energy network occur during distinct phases of a female's reproductive life (2), highlighting the potential benefits of creatine supplementation under specific conditions, such as menopause.

Menopause and the perimenopausal transition are marked by significant endocrine alterations, most notably a progressive decline in circulating estrogen levels. These hormonal changes have widespread effects across multiple organ systems, including the brain. In addition, the perimenopausal period is often accompanied by pronounced day-to-day fluctuations in estrogen concentrations, contributing to the variability of physiological and neuropsychological symptoms during this phase. Estrogen plays a critical role in modulating cerebral blood flow, mitochondrial function, synaptic

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plasticity, and neuroprotection. The reduction in estrogen during this transitional period is associated with neurophysiological alterations, such as reduced glucose metabolism in specific brain regions, impaired mitochondrial bioenergetics, and increased vulnerability to oxidative stress and neuroinflammation (3,4). These changes may contribute to commonly reported cognitive disturbances during perimenopause and menopause, including memory lapses, reduced attention, and executive dysfunction, collectively described as brain fog (5,6). Additionally, structural neuroimaging studies have revealed changes in brain morphology during the menopausal transition, including hippocampal volume reduction and cortical thinning, that might underlie the emergence or exacerbation of cognitive complaints, such as memory lapses, reduced processing speed, and impaired executive function (7).

Given its established role in cellular energy metabolism and emerging neuroprotective properties, creatine has been proposed as a potential intervention to mitigate brain-related changes associated with estrogen deficiency. Preclinical evidence indicates that creatine supplementation can enhance mitochondrial function, reduce oxidative damage, and support synaptic integrity under conditions of hormonal fluctuation and metabolic stress (8–10). However, evidence supporting the efficacy of creatine alone in perimenopausal women remains limited, as most preclinical and clinical studies have primarily investigated its effects in conjunction with resistance training (11). A recent population-based cross-sectional study demonstrated that optimal dietary creatine intake (~750 mg per day) was significantly associated with reduced risks of reproductive health issues in U.S. women, including perimenopausal conditions such as irregular menstrual cycles, pelvic pathology, and hormone replacement therapy use (12).

Despite these promising associations, the potential effects of creatine supplementation on perimenopausal and menopausal outcomes remain largely understudied—particularly in the context of low-dose formulations such as creatine hydrochloride and creatine ethyl ester. These alternative forms may exhibit enhanced solubility and bioavailability while reducing the risk of gastrointestinal discomfort commonly associated with higher doses of creatine monohydrate, the most extensively studied form of creatine supplementation. Preliminary evidence suggests that low-dose creatine formulations may confer beneficial effects across various conditions (13,14), and may elicit outcomes comparable to those observed with substantially higher doses of other creatine compounds (15). Furthermore, it remains unknown whether the co-administration of low-dose creatine salts with distinct physicochemical properties produces synergistic effects that enhance overall creatine uptake and tissue distribution. Consequently, the primary objective of this trial was to investigate the effects of medium-term supplementation with varying doses of creatine hydrochloride, either alone or in combination with creatine ethyl ester, on cognitive function, clinical outcomes, brain creatine levels, and biochemical markers in perimenopausal and menopausal women. We hypothesized that supplementation with creatine hydrochloride, particularly in combination with creatine ethyl ester, would lead to more substantial improvements in cognitive performance and perimenopausal symptoms and greater increases in brain creatine concentrations compared to placebo, in perimenopausal and menopausal women.

Methods

Trial design

The present study employed a parallel-group, randomized, placebo-controlled, double-blind design to evaluate the effects of low-dose creatine supplementation on neurocognitive and menopausal-related outcomes. Participants were randomly assigned in a 1:1:1:1 ratio to one of three experimental arms (see Experimental Intervention) or a placebo control group, using a computer-generated block randomization sequence. No modifications to the study design were made after the trial commenced. Allocation concealment was maintained using opaque, sealed envelopes, and both participants and investigators were blinded to group assignments throughout the intervention period. The study protocol was prospectively registered with ClinicalTrials.gov (NCT06660004), and no amendments to the trial design were made following trial initiation.

Participants

Eligibility criteria included apparently healthy perimenopausal and menopausal women being aged 40 years or older, having a body mass index (BMI) between 18.5 and 29.9 kg/m^2, and being free from Alzheimer's disease and dementia, as indicated by a Mini-Mental State Examination (MMSE) score of ≥25. Menopausal women were defined as having experienced no menstrual cycle for 12 consecutive months without other identifiable causes, while perimenopausal women were still menstruating (regularly or irregularly) but reported at least one perimenopausal symptom, such as hot flashes, sleep disturbances, mood swings, or concentration difficulties. Exclusion criteria encompassed the presence of major chronic diseases or acute medical conditions, use of dietary supplements within four weeks preceding enrollment, current or recent hormone replacement therapy, diagnosed psychiatric disorders or the use of prescription medications for such conditions, pregnancy or intention to become pregnant during the study period, abnormal clinical chemistry laboratory values, unwillingness or inability to attend follow-up assessments, and concurrent participation in other clinical trials. Prior to enrollment, all participants underwent a comprehensive pre-participation health screening, which included a health history assessment, general medical examination, and laboratory evaluations. Eligible participants voluntarily provided written informed consent to participate in this trial. The study protocol received ethical approval from the Institutional Review Board at the University of Novi Sad (# 50-08-14/2024-1) and was conducted in accordance with the ethical principles outlined in the Declaration of Helsinki (7th revision) and the International Council for Harmonization of Technical Requirements for Pharmaceuticals for Human Use (ICH) guidelines (https://www.ich.org/).

Experimental intervention

Participants were randomly allocated to one of four intervention products: low-dose creatine hydrochloride (L-CrHCL; 750 mg/day), medium-dose creatine hydrochloride (M-CrHCL; 1,500 mg/day), a combination of creatine hydrochloride and creatine ethyl ester (CrHCL-CEE; 800 mg/day, with 400 mg of each component), or a placebo (control group, CON; 513 mg/day of non-starch polysaccharides). The selected dosages and intervention duration were based on established risk-free protocols previously administered in various conditions for low-dose creatine forms (14,15). Participants consumed the interventions in capsule form, taken during breakfast and dinner with 200 mL of water. All interventions were identical in appearance, texture, and sensory characteristics. The intervention products were supplied at no cost by Vireo Systems (Nashville, TN). The intervention lasted eight weeks, during which participants were instructed to maintain their usual diet and physical activity, avoid the use of other dietary supplements, and refrain from making significant lifestyle changes.

Study outcomes

A list of prespecified primary and secondary outcome measures included cognitive function, clinical outcomes (five dimensions of fatigue, exercise tolerance, participant-reported outcomes), brain creatine levels, clinical chemistry panels, and the prevalence and severity of side effects. Measurements were conducted at two time points: baseline (pre-administration) and an 8-week follow-up (post-administration). The primary outcome was the change in mental fatigue scores—classified under the clinical outcomes domain (see below)—from baseline to the 8-week follow-up. Laboratory assessments were performed between 08:00 and 12:00, following an overnight fast.

Cognitive function

Cognitive function was evaluated using three computerized neuropsychological tests: The Alertness and Attention Network Test (ANT), the Symbol Digit Modalities Test (SDMT), and the Stroop Color and Word Test (SCWT) on the online testing platform Inquisit 6 (Millisecond Software, Seattle, WA, USA). The ANT assessed alerting, orienting, and executive attention networks within a unified task (16). The SDMT was employed to measure information processing speed (17), and the SCWT assessed cognitive interference inhibition, with stimuli classified as control, congruent, or incongruent (18).

Clinical outcomes

Fatigue was measured using the pencil-and-paper Multidimensional Fatigue Inventory (MFI-20) test (19), which evaluated five fatigue dimensions: general fatigue, physical fatigue, reduced motivation, reduced activity, and mental fatigue. Each item is rated on a 5-point Likert scale, yielding subscale scores ranging from 4 to 20, with higher scores indicating greater perceived fatigue. Exercise tolerance was assessed via an incremental treadmill test to exhaustion (TTE, sec), starting at 1.7 mph with a 10% gradient and progressing every three minutes to a maximum of 6 mph at a 22% gradient. Participant-reported outcomes for menopause-related symptoms (e.g., night sweats, hot flashes, depression, anxiety, mood swings, concentration difficulties, insomnia, headaches, muscle aches) were assessed using Visual Analog Scale (VAS) scale (20). For each symptom, participants indicated the perceived severity experienced over the preceding week by marking a point along a 10-cm horizontal VAS, anchored at 0 cm ("no symptoms") on the left and 10 cm ("most severe imaginable") on the right. The distance in centimeters and millimeters from the left anchor to the participant's mark was recorded as the symptom score, yielding a continuous and sensitive measure of symptom intensity. This approach enabled individualized and subjective assessment of symptom burden and was administered at both baseline and post-intervention assessments under standardized conditions to ensure consistency and reliability of reporting.

Brain creatine concentrations

Total brain creatine (tCr) concentrations were assessed in a subset of 16 participants, randomly selected to include four individuals from each of the four study groups. Quantification was performed using proton magnetic resonance spectroscopy (^{1}H-MRS) on a 1.5 Tesla Avanto scanner (Siemens, Erlangen, Germany) equipped with a matrix head coil operated in circularly polarized mode. Spectral data were acquired from anatomically predefined regions of interest, including the thalamus, frontal cortex, precentral gyrus, paracentral lobule, parietal cortex, and mesial parietal white and gray matter. These regions were selected based on their relevance to cognitive and metabolic functions potentially affected during the menopausal transition. Metabolite concentrations were analyzed using established post-processing protocols and quantification algorithms as previously described by (21), ensuring consistency in data extraction and metabolite ratio estimation. All scans were conducted by trained personnel under standardized conditions, and quality control measures were applied to ensure spectral integrity and reproducibility.

Clinical chemistry

At each lab visit, fasting blood samples were collected from the antecubital vein using gel vacutainers, centrifuged at 3,000 *g* within 10 min, and the serum was stored at −80 °C for biochemical and safety analyses. Serum concentrations of follicle-stimulating hormone (FSH), luteinizing hormone (LH), interleukin-6 (IL-6), tumor necrosis factor-alpha (TNF-α), and amyloid-β were quantified using commercial enzyme-linked immunosorbent assay (ELISA) kits from Elabscience Biotechnology (Wuhan, China); estradiol levels were measured using an ELISA kit from ELK Biotechnology (Denver, CO). Guanidinoacetic acid (GAA), creatine, and creatinine levels were analyzed using a modified liquid chromatography–tandem mass spectrometry method (Agilent 1200 Series LC System, Agilent Technologies Inc., Santa Clara, CA) as described previously (22). Safety panel markers, including C-reactive protein (CRP), clinical enzymes,

lipid profiles, and glucose levels, were analyzed using standard enzymatic methods on an automated analyzer (Hitachi 912, Tokyo, Japan).

Side effects

Adverse effects were monitored throughout the intervention period using a structured, open-ended questionnaire designed to capture both expected and unexpected events. Participants were specifically encouraged to report any gastrointestinal disturbances (e.g., stomach discomfort, constipation, diarrhea, nausea, vomiting) or systemic symptoms (e.g., headache, palpitations, dizziness), as well as any other physical or psychological complaints that may have emerged during the study. This questionnaire was administered on a weekly basis via scheduled telephone interviews conducted by trained study personnel, allowing for real-time reporting and clarification of symptoms. The weekly format facilitated consistent monitoring and enabled early identification of potential safety concerns. All reported adverse events were documented, categorized according to severity and duration, and evaluated for their potential relationship to the intervention. Where appropriate, additional follow-up was conducted to assess symptom resolution or progression. This approach ensured comprehensive and participant-centered safety surveillance throughout the study. Early termination criteria included severe side effects (e.g., prolonged vomiting or diarrhea) or significant health changes unrelated to the intervention. Compliance was monitored by capsule usage, and participants were considered compliant if they consumed more than 75% of the prescribed doses. All outcome measures were administered by trained personnel using standardized protocols. Assessment sessions were scheduled at consistent times for each participant to control for diurnal variability. Testing environments were kept consistent in terms of temperature, lighting, and noise levels. All instruments were calibrated prior to data collection, and inter-rater reliability was verified for subjective scales.

Statistical analyses

The minimum required sample size ($n = 28$) was calculated using G*Power 3.1 (Heinrich Heine University Düsseldorf, Germany), based on an effect size of 0.35 (small-to-medium effect), an alpha error probability of 0.05, statistical power of 0.80, four groups, and two measurements to assess changes in mental fatigue scores from baseline to post-intervention. To account for potential participant attrition, the sample size was increased by 25%. Initially, the data were analyzed using the Shapiro–Wilk test to evaluate normality and Bartlett's test to assess homogeneity of variance. The within-group differences during the trial were compared with paired t test for normally distributed data, and Wilcoxon Signed-Ranks Test for data not normally distributed, as appropriate. Interaction effects between time and intervention were analyzed using a two-way mixed-model ANOVA for data that satisfied the assumptions of normality and homogeneity of variance. Where significant interaction effects were observed, post hoc pairwise comparisons were conducted using Bonferroni correction to control for Type I error. For data with non-homogeneous variances, Friedman's two-way ANOVA by ranks was employed. In cases of significant Friedman test results, Wilcoxon signed-rank tests with Bonferroni-adjusted p-values were used for pairwise comparisons to identify specific within-group changes over time. The significance level was set at $p \leq 0.05$. All statistical analyses were conducted using SPSS version 24.0 for Mac (IBM SPSS Statistics, Chicago, IL, USA).

Table 1. Baseline Demographic and Clinical Characteristics of the Study Participants ($n = 36$)

Variables	
Age, years, mean ± SD	50.1 ± 5.7
Weight, kg, mean ± SD	66.7 ± 9.8
Body mass index, kg/m^2, mean ± SD	23.9 ± 3.0
MSSE score, points, mean ± SD	29.5 ± 0.7
Menopausal, n (%)	18 (50.0)

Results

A total of 36 apparently healthy perimenopausal and menopausal women were randomly assigned to one of the study groups, received the allocated intervention and were included in the primary analysis. The baseline demographic and clinical characteristics of the study cohort are summarized in Table 1.

The flow of participants throughout the trial, including enrollment, randomization, intervention allocation, follow-up, and analysis, is depicted in Figure 1.

Cognitive function

The changes in cognitive function are summarized in Table 2. L-CRHCL significantly enhanced alerting, reduced executive control deficits and reduced reaction time ($p \leq 0.05$), improved SDMT performance ($p = 0.02$), and decreased incongruent and overall reaction times as assessed by the SCWT test ($p \leq 0.05$) compared to baseline. Similarly, M-CRHCL led to a significant reduction in reaction time, incongruent and overall reaction times ($p \leq 0.05$), while CRHCL-CEE significantly improved alerting and decreased incongruent and overall reaction time at the 8-week follow-up ($p \leq 0.05$). In contrast, the control intervention did not elicit any significant changes in cognitive function biomarkers, except for a decreased incongruent and overall reaction times ($p \leq 0.05$). A two-way ANOVA revealed a significant interaction effect between the interventions on alertness, executive control, and reaction time ($p \leq 0.05$); experimental interventions demonstrated greater efficacy in improving these outcomes compared to the control intervention, with L-CRHCL and CRHCL-CEE showing superior effects on alertness, and L-CRHCL outperforming the control for both executive control and reaction time. A significant interaction effect was also observed for SDMT performance ($p < 0.01$), with participants in the L-CRHCL group exhibiting a significantly greater improvement in symbol recognition compared to those in other intervention groups. No

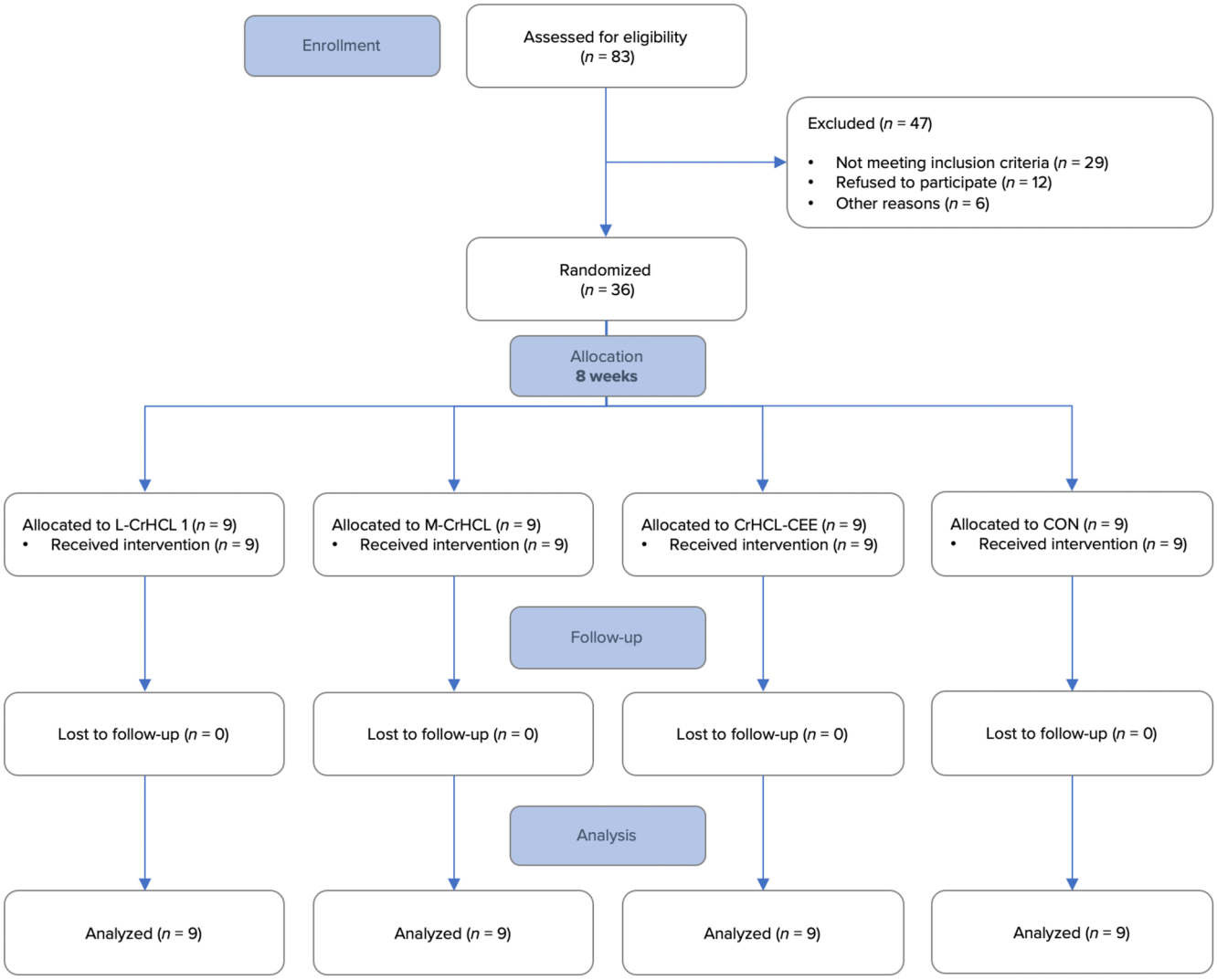

Figure 1. Participants flow throughout the trial. L-CrHCL = low-dose creatine hydrochloride (750 mg of creatine hydrochloride per day), MCrHCL = medium-dose creatine hydrochloride (1,500 mg of creatine hydrochloride per day), CrHCL-CEE = a combination of creatine hydrochloride and creatine ethyl ester (800 mg of creatine hydrochloride creatine plus creatine ethyl ester per day), CON = Control Group (513 mg of non-starch polysaccharides per day).

significant interaction effects were observed for the Stroop Color and Word Test outcomes throughout the study ($p > 0.05$).

Clinical outcomes

The changes in clinical outcomes related to fatigue, exercise tolerance, and participant-reported symptoms are presented in Table 3. Both M-CRHCL and CRHCL-CEE resulted in a significant reduction in general fatigue compared to baseline ($p \leq 0.05$). Furthermore, M-CRHCL led to a significant decrease in mood swings and concentration difficulties ($p \leq 0.05$), while CRHCL-CEE significantly reduced anxiety at follow-up ($p = 0.01$). In contrast, neither the control intervention nor L-CRHCL elicited significant changes in clinical outcomes ($p > 0.05$). Additionally, a notable trend toward an interaction effect between treatment and time was observed for mood swings ($p = 0.06$).

Brain creatine concentrations

The changes in brain creatine levels are presented in Table 4. L-CRHCL induced a significant increase in tCr levels across the majority of evaluated regions (nine out of thirteen locations) ($p \leq 0.05$), including the left and right frontal white matter, left and right frontal grey matter, right precentral white matter, left and right paracentral grey matter, and left and right parietal mesial grey matter compared to baseline. M-CRHCL led to a significant increase in tCr levels in five out of thirteen assessed regions, including the thalamus, left and right paracentral grey matter, and left and right parietal mesial grey matter ($p \leq 0.05$). CRHCL-CEE significantly increased brain creatine levels at follow-up in the thalamus, left frontal white matter, right frontal grey matter, right precentral white matter, and left and right parietal mesial grey matter ($p \leq 0.05$). In contrast, the control intervention resulted in a significant reduction in brain creatine levels at follow-up in the left frontal white matter and right frontal grey matter, while increasing total creatine in the right

Table 2. Cognitive Function During the Trial

	Control group ($n=9$)		Low-dose creatine HCl ($n=9$)		Medium-dose creatine HCl ($n=9$)		Creatine HCl + creatine ethyl ester ($n=9$)		
	Baseline	Follow-up	Baseline	Follow-up	Baseline	Follow-up	Baseline	Follow-up	p*
Attention Network Test									
Alerting (score)	28.1 ± 20.5	44.2 ± 43.4	10.7 ± 26.5	40.7 ± 35.1†	30.4 ± 27.7	27.7 ± 34.0	1.4 ± 25.5	42.4 ± 27.6†	0.03
Orienting (score)	31.1 ± 30.7	28.4 ± 27.4	43.4 ± 29.2	38.0 ± 36.5	27.4 ± 38.0	17.2 ± 38.6	40.2 ± 54.6	21.7 ± 27.7	0.90
Executive control (score)	122.3 ± 21.0	114.1 ± 29.8	133.2 ± 32.2	111.8 ± 25.3†	124.8 ± 53.2	118.9 ± 64.2	124.9 ± 45.2	98.6 ± 23.0	0.03
Reaction time (ms)	651 ± 54	659 ± 82	661 ± 86	622 ± 89†	726 ± 106	678 ± 1059†	748 ± 100	717 ± 95	0.02
Test accuracy (%)	98.7 ± 1.6	99.0 ± 1.3	98.6 ± 1.5	99.0 ± 1.4	99.2 ± 1.3	99.5 ± 0.6	99.1 ± 0.9	99.4 ± 0.7	0.71
Symbol Digit Modalities Test									
Correct symbols (score)	56.4 ± 12.7	62.8 ± 16.1	53.6 ± 7.9	58.7 ± 8.7†	47.9 ± 6.6	50.0 ± 5.1	46.4 ± 10.9	50.4 ± 11.1	<0.01
Errors (score)	1.1 ± 1.7	1.9 ± 3.2	0.8 ± 1.1	1.3 ± 2.2	0.4 ± 0.7	0.9 ± 1.3	1.8 ± 2.9	1.6 ± 2.0	0.87
Stroop Color and Word Test									
Congruent reaction time (ms)	555 ± 62	565 ± 75	565 ± 57	544 ± 50	609 ± 58	580 ± 52	633 ± 101	621 ± 112	0.23
Incongruent reaction time (ms)	943 ± 175	858 ± 166†	918 ± 204	794 ± 89†	997 ± 138	899 ± 135†	1143 ± 288	1027 ± 254†	0.90
Control reaction time (ms)	440 ± 52	437 ± 53	441 ± 24	4222 ± 43	457 ± 23	438 ± 40	467 ± 68	456 ± 70	0.30
Overall reaction time (ms)	646 ± 86	620 ± 91†	641 ± 78	586 ± 55†	688 ± 61	639 ± 60†	748 ± 131	701 ± 130†	0.49
Congruent accuracy (%)	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	0.99
Incongruent accuracy (%)	98.4 ± 1.5	98.7 ± 1.6	98.4 ± 1.3	99.1 ± 0.8	98.7 ± 1.0	99.1 ± 0.9	97.1 ± 3.2	98.8 ± 1.2	0.80
Control accuracy (%)	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	0.99
Overall accuracy (%)	99.5 ± 0.5	99.6 ± 0.5	99.5 ± 0.4	99.7 ± 0.3	99.6 ± 0.3	99.7 ± 0.3	99.1 ± 1.1	99.6 ± 0.4	0.80

Values are mean ± SD.

Notes: Cross (†) indicates significant difference at $p \leq 0.05$ for within-group comparisons versus baseline values. Asterisk (*) indicates statistical significance for mixed model ANOVA interaction effect or Friedman's two-way ANOVA by ranks where appropriate.

paracentral grey matter ($p \leq 0.05$). A Friedman's two-way ANOVA by ranks revealed a significant interaction effect between interventions ($p = 0.01$), with L-CRHCL and CRHCL-CEE demonstrating superiority over control group in augmenting total creatine levels in the right frontal white matter. Similarly, a significant interaction effect was observed for the left frontal grey matter ($p = 0.02$), where L-CRHCL and CRHCL-CEE were superior to control group in affecting total creatine levels in this region. In addition, a non-significant interaction between interventions was reported for several other locations ($p < 0.20$), including right frontal grey matter, left precentral white matter, left and right paracentral white matter, left and right parietal white matter, and right parietal mesial grey matter.

Clinical chemistry

The changes in clinical chemistry profiles are summarized in Table 5. No significant changes were observed in relevant hormonal biomarkers and inflammatory cytokines throughout the study, except for a significant reduction in serum amyloid-β levels in the control group at the 8-week follow-up compared to baseline ($p = 0.02$). No significant interaction effects were detected for any of the variables; however, a notable trend toward an interaction effect was observed for serum amyloid-β ($p = 0.08$), indicating a potential pathophysiological disadvantage of the placebo in reducing this biomarker, whereas levels exhibited a slight increase in the other groups. Serum IL-6 levels remained undetectable (below the detection limit of 1.56 pg/mL) in over 90% of participants both before and after the intervention and were therefore excluded from within- and between-group comparisons.

No significant changes were observed in circulating GAA, creatine, or creatinine levels over the course of the study. However, a Friedman's two-way ANOVA by ranks revealed a significant interaction effect for serum creatine levels ($p = 0.03$).

In addition, no major alterations in safety panel markers, including CRP, clinical enzymes, lipid profiles, and glucose levels, were observed throughout the trial. However, a significant reduction in serum creatine kinase (CK) levels was noted in the control group ($p = 0.04$), while serum glucose levels significantly decreased in the L-CRHCL group ($p = 0.02$) compared to baseline values. Notably, a significant interaction effect was observed for changes in low-density lipoprotein cholesterol ($p = 0.01$) and triglyceride levels ($p = 0.04$), with CRHCL-CEE exhibiting a superior capacity to reduce these biomarkers compared to other groups, in which levels either increased or remained unchanged.

Side effects of intervention

All participants who completed the trial did not report any major side effects or adverse events that interfered with their participation. Following the administration of L-CRHCL, three participants experienced mild, transient episodes of heartburn, while one participant reported increased energy levels. In the M-CRHCL group, one participant noted an improvement in overall well-being. In the control group, one participant reported bloating, which resolved after the first four weeks of administration, and two participants reported feeling stronger and more toned. Body mass remained stable across all intervention groups over the course of the study ($p = 0.50$). Specifically, no significant within- or between-group differences were observed: L-CRHCL group (baseline: 68.1 ± 10.2 kg; follow-up: 68.5 ± 10.3 kg), M-CRHCL group (baseline: 67.1 ± 8.3 kg; follow-up: 67.1 ± 8.2 kg), CRHCL+CEE group (baseline: 68.9 ± 11.8 kg; follow-up: 69.0 ± 12.5 kg), and control group (baseline: 67.1 ± 8.9 kg; follow-up: 67.1 ± 9.9 kg). Compliance with the intervention was high across all groups, with adherence rates of 94.8 ± 5.1% in L-CRHCL, 93.0 ± 7.2% in M-CRHCL, 95.7 ± 2.6% in CRHCL-CEE, and 94.4 ± 3.7% in the control group ($p = 0.70$).

Table 3. Clinical Outcomes During the Trial

	Control group ($n=9$)		Low-dose creatine HCl ($n=9$)		Medium-dose creatine HCl ($n=9$)		Creatine HCl + creatine ethyl ester ($n=9$)		
	Baseline	Follow-up	Baseline	Follow-up	Baseline	Follow-up	Baseline	Follow-up	p*
Fatigue									
General fatigue (points)	9.6 ± 3.5	9.2 ± 3.1	10.6 ± 3.4	8.3 ± 2.1	11.2 ± 1.0	9.6 ± 1.7†	11.3 ± 5.1	9.1 ± 3.2†	0.47
Physical fatigue (points)	7.8 ± 4.8	7.4 ± 2.8	7.4 ± 4.1	7.0 ± 2.2	9.6 ± 2.8	9.2 ± 2.3	11.6 ± 5.1	9.9 ± 4.7	0.16
Reduced motivation (points)	8.4 ± 4.0	7.3 ± 2.2	7.7 ± 3.1	7.4 ± 1.9	8.1 ± 1.7	6.8 ± 1.7	7.2 ± 2.9	9.2 ± 4.1	0.86
Reduced activity (points)	8.1 ± 4.6	6.3 ± 2.4	7.3 ± 2.6	7.9 ± 2.5	8.6 ± 2.7	7.9 ± 1.8	8.2 ± 4.4	9.9 ± 4.6	0.73
Mental fatigue (points)	9.4 ± 3.5	9.1 ± 2.0	8.8 ± 3.7	8.3 ± 3.2	9.9 ± 3.5	8.8 ± 4.0	9.4 ± 3.0	9.1 ± 2.1	0.92
Time to exhaustion (sec)	868 ± 259	836 ± 244	786 ± 161	761 ± 159	701 ± 122	742 ± 133	723 ± 117	761 ± 135	0.42
Participant-reported symptoms									
Night sweats (cm)	1.0 ± 1.2	1.6 ± 1.6	1.6 ± 2.6	0.7 ± 0.9	2.1 ± 2.1	1.6 ± 1.7	0.9 ± 1.3	1.0 ± 1.4	0.87
Hot flashes (cm)	0.3 ± 0.6	0.5 ± 0.5	2.0 ± 2.5	1.6 ± 3.1	0.7 ± 1.0	0.6 ± 0.8	1.0 ± 1.9	0.4 ± 0.7	0.69
Depression (cm)	0.6 ± 0.9	1.1 ± 0.9	1.3 ± 1.8	1.5 ± 2.2	0.4 ± 0.4	0.3 ± 0.4	2.2 ± 2.7	1.7 ± 2.3	0.27
Anxiety (cm)	1.7 ± 1.8	1.6 ± 2.1	1.8 ± 2.7	2.1 ± 3.3	1.8 ± 2.5	0.8 ± 0.8	3.1 ± 2.5	1.9 ± 1.5†	0.73
Mood swings (cm)	1.5 ± 1.6	0.7 ± 0.8	2.3 ± 3.5	2.1 ± 2.9	2.4 ± 2.0	0.5 ± 0.7†	3.3 ± 3.1	2.4 ± 2.3	0.06
Concentration difficulties (cm)	1.4 ± 1.3	1.2 ± 1.4	2.8 ± 2.9	2.5 ± 2.5	2.5 ± 2.7	1.3 ± 1.8†	2.7 ± 3.0	2.5 ± 2.0	0.66
Insomnia (cm)	0.7 ± 1.2	0.8 ± 1.2	3.1 ± 4.3	2.4 ± 3.0	1.4 ± 1.3	0.7 ± 0.8	2.9 ± 2.9	1.6 ± 2.5	0.57
Headaches (cm)	1.3 ± 1.6	1.6 ± 1.4	2.9 ± 4.0	2.7 ± 4.0	3.0 ± 3.9	3.0 ± 3.9	1.6 ± 3.1	1.2 ± 1.8	0.73
Muscle aches (cm)	0.5 =- 1.2	0.4 ± 0.5	0.6 ± 1.4	0.6 ± 0.9	1.7 ± 2.6	1.1 ± 1.6	2.8 ± 3.4	1.8 ± 2.5	0.38

Values are mean ± SD.

Notes: Cross (†) indicates significant difference at $p \leq 0.05$ for within-group comparisons versus baseline values. Asterisk (*) indicates statistical significance for mixed model ANOVA interaction effect or Friedman's two-way ANOVA by ranks where appropriate.

Table 4. Total Brain Creatine Levels (mM) During the Trial

	Control group ($n=4$)		Low-dose creatine HCl ($n=4$)		Medium-dose creatine HCl ($n=4$)		Creatine HCl + creatine ethyl ester ($n=4$)		
	Baseline	Follow-up	Baseline	Follow-up	Baseline	Follow-up	Baseline	Follow-up	p*
Thalamus	5.88 ± 1.11	5.85 ± 0.76	5.79 ± 0.65	6.33 ± 0.69	5.07 ± 0.94	6.36 ± 0.44†	5.29 ± 0.78	6.26 ± 0.42†	0.47
Left frontal WM	5.83 ± 0.73	5.60 ± 0.74†	6.47 ± 0.87	7.09 ± 0.73†	6.64 ± 0.34	6.96 ± 0.34	6.72 ± 0.32	7.55 ± 0.60†	0.41
Right frontal WM	5.38 ± 0.38	5.43 ± 0.49	6.27 ± 0.64	7.15 ± 0.54†	5.98 ± 1.15	6.96 ± 0.67	6.48 ± 0.63	7.84 ± 0.40	0.01
Left frontal GM	5.95 ± 0.71	5.90 ± 1.13	6.99 ± 0.49	7.94 ± 0.53†	6.60 ± 0.95	7.14 ± 0.49	5.91 ± 0.52	7.54 ± 0.81	0.02
Right frontal GM	6.58 ± 0.75	6.30 ± 0.90†	6.64 ± 0.64	8.19 ± 0.65†	6.31 ± 1.41	7.08 ± 0.49	5.59 ± 1.19	8.16 ± 0.63†	0.12
Left precentral WM	5.83 ± 0.44	5.80 ± 0.42	6.01 ± 0.81	6.27 ± 0.46	6.17 ± 0.58	6.84 ± 0.26	6.20 ± 0.71	6.83 ± 0.75	0.13
Right precentral WM	5.73 ± 0.36	5.90 ± 0.34	6.13 ± 1.03	6.49 ± 1.18†	6.04 ± 0.35	6.31 = −0.67	5.70 ± 0.77	6.73 ± 0.66†	0.29
Left paracentral GM	6.58 ± 0.48	6.43 ± 0.67	6.69 ± 1.69	8.43 ± 1.39†	6.79 ± 0.88	8.14 ± 1.26†	6.09 ± 1.36	7.01 ± 1.71	0.16
Right paracentral GM	6.53 ± 0.33	6.83 ± 0.49†	7.21 ± 1.55	8.22 ± 1.23†	7.09 ± 0.57	7.78 ± 0.72†	6.93 ± 0.54	7.32 ± 0.63	0.13
Left parietal WM	5.38 ± 0.40	5.13 ± 0.51	5.11 ± 1.29	5.74 ± 0.85	5.71 ± 0.55	6.05 ± 0.27	5.63 ± 1.05	6.56 ± 1.53	0.14
Right parietal WM	5.20 ± 0.75	5.18 ± 0.96	5.80 ± 0.96	6.96 ± 0.79	6.15 ± 1.19	5.95 ± 0.80	5.93 ± 0.94	6.26 ± 1.16	0.10
Left parietal mesial GM	6.78 ± 0.76	6.68 ± 0.92	7.16 ± 0.66	8.79 ± 1.13†	6.56 ± 0.84	7.69 ± 1.22†	6.87 ± 0.61	8.26 ± 0.46†	0.33
Right parietal mesial GM	7.08 ± 0.40	7.28 ± 0.30	7.60 ± 0.14	8.75 ± 0.83†	6.88 ± 0.70	8.13 ± 1.34†	7.10 ± 0.95	8.84 ± 0.93†	0.16

Values are mean ± SD. *Abbreviations*: WM = white matter, GM = grey matter. Cross (†) indicates significant difference at $p \leq 0.05$ for within-group comparisons versus baseline values. Asterisk (*) indicates statistical significance for mixed model ANOVA interaction effect or Friedman's two-way ANOVA by ranks where appropriate.

Discussion

To the best of our knowledge, this trial is the first to investigate the effects of low-dose creatine formulations on clinical outcomes and relevant biomarkers in perimenopausal and menopausal women. Our findings indicate that an eight-week regimen of creatine hydrochloride (with or without creatine ethyl ester) was superior to placebo in enhancing cognition, increasing frontal brain creatine levels, and favorably modulating serum lipid profiles in this population. Additionally, creatine hydrochloride (1,500 mg; M-CRHCL) demonstrated a potential advantage over other interventions in mitigating the severity of mood swings ($p = 0.06$). Daily supplementation with medium-dose creatine hydrochloride (M-CRHCL) also resulted in a significant reduction in general fatigue and concentration difficulties, while the combination of creatine hydrochloride and creatine ethyl ester (CRHCL-CEE) was associated with a notable alleviation of anxiety symptoms at the eight-week follow-up. Importantly, all interventions were well tolerated, with no reported severe adverse effects and a 0% attrition rate. Interestingly, body mass remained stable throughout the intervention period, with no significant changes observed within or between groups. This finding suggests that low-dose creatine supplementation does not promote weight gain in this cohort, potentially alleviating a common concern and reducing barriers to creatine use among women. These findings suggest that this supplementation protocol holds promise as a safe, effective and practical dietary strategy for improving clinical outcomes and elevating brain creatine concentrations in perimenopausal and menopausal women. Further research is warranted to elucidate the underlying mechanisms and long-term benefits of creatine supplementation in this demographic.

Creatine is increasingly recognized as a beneficial nutrient for women's health, including its potential role in supporting well-being during menopause. A recent review highlighted that creatine supplementation, most commonly consumed as creatine monohydrate at doses of 3–5 grams per day, offers significant benefits for improving exercise performance, muscle and bone health, and recovery in both

Table 5. Clinical Chemistry Profiles During the Trial

	Control group ($n=9$)		Low-dose creatine HCl ($n=9$)		Medium-dose creatine HCl ($n=9$)		Creatine HCl + creatine ethyl ester ($n=9$)		
	Baseline	Follow-up	Baseline	Follow-up	Baseline	Follow-up	Baseline	Follow-up	P^*
FSH (mIU/mL)	4.0 ± 3.2	3.6 ± 2.9	4.8 ± 2.9	5.3 ± 4.2	6.6 ± 6.2	4.1 ± 1.8	2.7 ± 1.6	2.9 ± 0.2	0.66
LH (mIU/mL)	8.7 ± 9.3	11.2 ± 10.7	13.7 ± 6.9	16.8 ± 9.3	11.8 ± 9.6	10.6 ± 7.6	14.3 ± 18.2	15.8 ± 12.3	0.97
Estradiol (pg/mL)	58.9 ± 48.5	70.8 ± 69.1	61.0 ± 85.4	113.6 ± 145.5	136.4 ± 104.6	139.4 ± 205.3	56.1 ± 60.0	46.1 ± 65.4	0.42
IL-6 (pg/mL)	<1.56	<1.56	<1.56	<1.56	<1.56	<1.56	<1.56	<1.56	–
TNF-α (pg/mL)	174.7 ± 199.3	121.9 ± 156.4	80.5 ± 159.8	145.2 ± 208.2	134.7 ± 209.9	127.9 ± 207.2	138.9 ± 206.6	138.5 ± 205.8	0.48
Amyloid-β (pg/mL)	899.7 ± 838.7	786.0 ± 851.0†	1048.5 ± 533.3	1081.5 ± 559.8	1382.4 ± 867.9	1418.5 ± 918.0	1476.8 ± 940.7	1403.4 ± 910.8	0.08
GAA (μmol/L)	2.8 ± 0.9	2.8 ± 0.5	2.7 ± 0.5	2.8 ± 0.7	2.5 ± 1.0	2.9 ± 1.5	2.7 ± 0.3	2.4 ± 0.5	0.74
Creatine (μmol/L)	37.2 ± 17.4	46.4 ± 22.6	39.8 ± 24.9	57.6 ± 20.7	41.9 ± 28.8	102.8 ± 130.8	37.4 ± 21.4	43.3 ± 21.1	0.03
Creatinine (μmol/L)	62.6 ± 9.0	67.2 ± 14.6	59.3 ± 8.1	69.2 ± 12.8	76.6 ± 39.2	61.9 ± 12.6	67.2 ± 7.2	64.5 ± 6.7	0.38
C-reactive protein (mg/L)	0.9 ± 0.4	0.7 ± 0.1	1.1 ± 1.0	1.3 ± 1.4	1.8 ± 1.5	1.2 ± 0.6	2.1 ± 2.3	1.0 ± 0.8	0.82
AST (IU/L)	23 ± 6	24 ± 10	25 ± 11	23 ± 8	22 ± 6	21 ± 4	21 ± 4	22 ± 4	0.87
ALT (IU/L)	23 ± 10	24 ± 7	25 ± 16	22 ± 9	23 ± 13	20 ± 8	21 ± 5	21 ± 8	0.69
GGT (IU/L)	16 ± 11	15 ± 7	17 ± 7	17 ± 8	14 ± 5	14 ± 4	15 ± 7	14 ± 6	0.73
ALP (IU/L)	55 ± 16	51 ± 12	63 ± 14	61 ± 15	78 ± 21	72 = −15	72 ± 20	68 ± 16	0.63
CK (IU/L)	106 ± 41	74 ± 20†	95 ± 38	124 ± 112	79 ± 23	81 ± 31	75 ± 27	84 ± 37	0.31
Glucose (mmol/L)	4.9 ± 0.3	4.8 ± 0.2	5.4 ± 0.9	5.1 ± 0.7†	5.0 ± 0.8	4.9 ± 0.4	4.9 ± 0.4	4.9 ± 0.7	0.52
Total cholesterol (mmol/L)	5.4 ± 1.2	5.5 ± 0.9	5.3 ± 0.9	5.3 ± 0.9	6.0 ± 0.7	6.0 ± 0.9	5.8 ± 0.8	5.6 ± 0.8	0.54
LDL-cholesterol (mmol/L)	3.1 ± 0.8	3.3 ± 0.8	2.8 ± 0.7	2.9 ± 0.7	3.6 ± 0.5	3.7 ± 0.8	3.5 ± 0.7	3.4 ± 0.7	0.01
HDL-cholesterol (mmol/L)	1.9 ± 0.4	1.8 ± 0.2	2.0 ± 0.6	1.9 ± 0.6	1.7 ± 0.2	1.6 ± 0.3	1.7 ± 0.3	1.7 ± 0.4	0.32
Triglycerides (mmol/L)	0.8 ± 0.2	0.9 ± 0.2	0.9 ± 0.3	1.0 ± 0.5	1.5 ± 0.9	1.5 ± 1.0	1.3 ± 0.7	1.1 ± 0.2	0.04

Values are mean ± SD. *Abbreviations*: FSH = follicle-stimulating hormone, LH = luteinizing hormone, IL6 = interleukin-6, TNF-α = tumor necrosis factor-alpha, GAA = guanidinoacetic acid, AST = aspartate transaminase, ALT = alanine transaminase, GGT = gamma-glutamyl transferase, ALP = alkaline phosphatase, CK = creatine kinase, LDL = low-density lipoprotein, HDL = high-density lipoprotein. Cross (†) indicates significant difference at $p \leq 0.05$ for within-group comparisons versus baseline values. Asterisk (*) indicates statistical significance for mixed model ANOVA interaction effect or Friedman's two-way ANOVA by ranks where appropriate.

premenopausal and menopausal women. However, these benefits appear to be primarily contingent upon concurrent physical activity (23). Even at high doses of creatine monohydrate (~ 20 grams per day), when administered without exercise interventions, the effects on muscle and bone health in postmenopausal women have been reported to be modest (for a detailed review, see (11). Additionally, creatine supplementation has shown promise as an adjunct therapeutic approach for females with depression, including those in the perimenopausal and menopausal stages (24–26). Still, the effects of creatine supplementation in women outside of exercise or pharmacological interventions remain underexplored. A recent population-based cross-sectional study identified a significant association between optimal dietary creatine intake (≥ 750 mg per day) and a lower risk of irregular menstrual cycles, obstetric complications, and pelvic pathologies (12). This study, which analyzed data from 4,522 females aged 12 years and older, including both perimenopausal and menopausal individuals, provides noteworthy evidence supporting the association between creatine intake and women's health. Our study builds upon previous research by further confirming the beneficial effects of creatine supplementation for perimenopausal and menopausal women. It expands the existing body of evidence by evaluating the effects of specific low-dose supplementation regimens without the inclusion of adjunct pharmacological treatments or structured exercise programs. Moreover, the present study provides a comprehensive assessment of cognitive function, brain creatine levels, (peri)menopause-related symptoms, and key biomarkers related to creatine turnover, hormonal balance, and pro-inflammatory activation, thereby offering novel insights into the role of creatine in women's health during this transitional stage.

Perimenopausal and postmenopausal women experience notable brain-related physiological changes, largely attributed to declining and fluctuating estrogen levels (27). These alterations include impaired cerebral glucose metabolism, increased oxidative stress, mitochondrial dysfunction, and reduced synaptic plasticity, which may contribute to cognitive complaints, mood disturbances, and reduced mental clarity commonly reported during this transition (28). Neuroimaging studies further support region-specific structural and functional changes, such as reduced hippocampal volume and altered neural connectivity (29). Given creatine's role in cellular energy buffering, neuroprotection, and mitochondrial support, it may offer a targeted strategy to mitigate some of these menopause-associated neurophysiological changes, particularly in brain regions sensitive to hormonal modulation and energy demand. Our trial demonstrated a significant effect of creatine supplementation on increasing brain creatine levels, with all three experimental interventions resulting in elevated total creatine concentrations across the majority of evaluated brain regions. Notably, the most pronounced effects were observed in the frontal gray and white matter, where creatine supplementation significantly outperformed placebo in elevating creatine concentrations. Creatine plays a crucial role as an energy-providing molecule for the brain and other metabolically demanding organs (30). Previous research indicates that females tend to have lower creatine levels in the brain, particularly in the frontal lobe (31), a region critical for mood regulation, cognition, memory, and emotional processing. Given these findings, creatine hydrochloride supplementation may constitute an effective strategy for supporting brain creatine levels in perimenopausal and menopausal women, with the observed increases potentially associated with enhancements in

cognitive function and clinical outcomes within this population. In our study, creatine hydrochloride demonstrated a robust effect on enhancing alerting, processing speed and reducing reaction time, along with a notable trend toward reducing mood swings. These benefits may be attributed not only to improved brain energy metabolism but also to additional neuroprotective effects of creatine, including its antioxidant and neuromodulatory properties (32). A potential neuroprotective effect of creatine is also suggested by the observed trend of preserved serum amyloid-β levels in all three experimental groups, whereas levels declined in the control group. Although serum amyloid-β levels exhibit considerable interindividual variability in the general population, independent of cognitive status, and lower levels are not necessarily detrimental (33), emerging evidence suggests that reduced circulating amyloid-β may be linked to metabolic dysregulation, increased cerebral amyloid deposition, and impaired neuronal plasticity (34). This relationship appears particularly relevant for individuals at heightened risk of cognitive decline (35), such as menopausal women. Given the potential implications for brain health, further research is needed to explore whether the addition of creatine to traditional interventions targeting amyloid-β metabolism—including hormone replacement therapy, dietary and lifestyle modifications, and pharmacological approaches—could help mitigate the risks associated with menopause-related neurobiological changes. The potential effects of creatine supplementation on mood swings should be interpreted with caution, given the influence of multiple confounding factors. Mood-related symptoms in perimenopausal and menopausal women are subject to substantial inter-day variability and may fluctuate independently of the intervention due to psychosocial, environmental, and hormonal influences (36,37). Additionally, the reproductive cycle—particularly in perimenopausal women with irregular but persisting menstrual activity—can introduce endogenous hormonal fluctuations that further affect mood regulation (38,39). These factors complicate the attribution of observed mood-related changes solely to creatine supplementation and highlight the need for more controlled studies with frequent and time-sensitive assessments of mood.

Interestingly, despite the relatively low creatine dosage used in this trial, we observed significant increases in brain creatine levels. This is in contrast to previous studies that typically employ higher doses of creatine monohydrate (e.g., 20 grams per day) for various neurological conditions (40). Several potential mechanisms may account for this discrepancy. These include: (1) menopause-related cerebrovascular and structural brain changes that may facilitate increased creatine uptake from the circulation; (2) a distinct transport mechanism that enhances the permeability of creatine hydrochloride—and potentially creatine ethyl ester—across the blood–brain barrier; (3) an intervention-induced upregulation of endogenous creatine synthesis within the brain; (4) reduced cerebral creatine clearance, resulting in prolonged retention; (5) synergistic effects among these mechanisms; and (6) other as-yet unidentified physiological processes that merit further investigation. Previous studies have demonstrated that brain delivery differs among various creatine compounds and analogs (41,42), with even minor structural differences potentially influencing their ability to cross into cerebral tissues. These observations highlight the need for further research to characterize the brain bioavailability and pharmacokinetics of creatine hydrochloride and creatine ethyl ester specifically. Additionally, no substantial differences were observed between the varying doses of creatine hydrochloride for most outcomes, suggesting the absence of a clear dose-response effect for the dosages employed in the current trial. This may be attributed to factors such as saturation of creatine transporters, a finite capacity for creatine storage and uptake, excretion of excess creatine, and possible reductions in endogenous creatine production. These mechanisms warrant further investigation through detailed pharmacokinetic and pharmacodynamic studies across different doses of creatine hydrochloride. Nevertheless, lower doses appeared to exert a greater impact on cognitive function and brain metabolism, while medium doses were more effective in improving clinical outcomes. Our findings suggest that creatine hydrochloride supplementation may offer a promising, targeted approach to improving brain creatine levels and cognitive function in perimenopausal and menopausal women, potentially addressing critical neurological and metabolic vulnerabilities associated with this life stage. Due to possible estrogen-driven differences in brain creatine concentrations, supplementation with creatine hydrochloride may be thus particularly effective in females for increasing brain creatine levels and promoting a pro-energetic neural environment.

Our study's methodology was robust due to its randomized, placebo-controlled, double-blind design; however, several limitations exist. The sample size, although justified through power analysis, remains small, potentially limiting generalizability. However, the interpretation of the findings would have been enhanced by a clearer distinction between perimenopausal and postmenopausal participants in relation to study outcomes, while stratified randomization according to menopausal status could have offered improved contextual insight into the clinical relevance of the observed effects. The reliance on self-reported outcomes for symptoms and compliance introduces bias, and the exclusion of women with major chronic diseases may reduce real-world applicability. The intervention duration of eight weeks may be insufficient to observe long-term cognitive and metabolic effects. We did not implement a weight-adjusted dosing strategy for creatine supplementation, resulting in considerable variability in relative creatine exposure. This discrepancy is particularly relevant given the substantial range in body mass across the sample (from 48.9 to 87.6 kg). The study also lacks dietary and physical activity monitoring beyond self-reported maintenance, which could confound results. Additionally, the assessment of brain creatine levels in only sixteen participants limits statistical power for this outcome. Finally, the use of a multi-component mixture consisting of creatine hydrochloride and creatine ethyl ester complicates the ability to delineate the specific contributions of each component to the observed effects for this intervention. To enhance the impact and reproducibility of findings, future studies should include larger, stratified samples by menopausal status (ideally paring with blood FSH levels),

adopt weight-adjusted dosing, and extend intervention duration. Incorporating objective measures of adherence, dietary intake, and physical activity would minimize bias, while assessing brain creatine in a larger cohort and evaluating individual supplement components separately would improve mechanistic insight and interpretation of effects.

Although the experimental design did not allow for the direct calculation of test-retest reliability within the present cohort, the cognitive, clinical, and neurochemical measures employed in this study are supported by well-established psychometric properties in the literature. For example, both the SDMT and the SCWT have demonstrated high test-retest reliability coefficients in comparable adult populations ($r = 0.80–0.95$) (43,44). Similarly, the MFI-20, used to assess mental and general fatigue, has shown strong internal consistency (Cronbach's $\alpha > 0.80$) and satisfactory test-retest reliability over a 1–2 week interval ($r = 0.76–0.85$) (19,45). Additionally, brain creatine concentrations measured by proton MRS can be quantified with high reliability, with reported within-subject variation typically below 2% (46). The use of these validated instruments supports the interpretability of the observed pre-to-post intervention changes, particularly in light of the significant between-group differences in cognitive performance, clinical symptoms, and brain bioenergetic profiles. Nevertheless, future studies should consider incorporating repeated baseline assessments to empirically establish within-sample test-retest reliability and further strengthen methodological rigor.

Conclusion

Taking creatine hydrochloride for eight weeks appears to enhance cognition and brain creatine levels, while also potentially attenuating the severity of mood swings in perimenopausal and menopausal women, likely due to its neuroprotective effects. These findings underscore the potential benefits of low-dose creatine formulations with superior solubility profiles in addressing common (peri) menopausal features. Further research in larger cohorts and with extended follow-up periods is needed to validate our results and clarify the underlying mechanisms driving these effects.

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Ethical approval

Study approval statement: The ethical approval was granted by the local Institutional Review Board (IRB) at the University of Novi Sad under protocol # 50-08-14/2024-1. Consent to participate statement: Written informed consent was obtained from all respondents to participate in the study. The research was conducted ethically following the World Medical Association Declaration of Helsinki.

Author contributions

CRediT: **Darinka Korovljev**: Data curation, Investigation, Methodology, Project administration, Supervision, Writing – review & editing; **Jelena Ostojic**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – review & editing; **Jovana Panic**: Data curation, Formal analysis, Investigation, Methodology, Writing – review & editing; **Marijana Ranisavljev**: Data curation, Formal analysis, Investigation, Methodology, Writing – review & editing; **Nikola Todorovic**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – review & editing; **David Nedeljkovic**: Data curation, Formal analysis, Investigation, Methodology, Writing – review & editing; **Jovan Kuzmanovic**: Data curation, Formal analysis, Investigation, Methodology, Writing – review & editing; **Milan Vranes**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Supervision, Writing – review & editing; **Valdemar Stajer**: Conceptualization, Data curation, Investigation, Methodology, Supervision, Writing – review & editing; **Sergej. M Ostojic**: Conceptualization, Data curation, Formal analysis, Funding acquisition, Project administration, Resources, Supervision, Writing – original draft.

Disclosure statement

SMO serves as a member of the Scientific Advisory Board on Creatine in Health and Medicine (AlzChem LLC). SMO co-owns patent "Supplements Based on Liquid Creatine" at the European Patent Office (WO2019150323 A1) and patent application "Composition Comprising Creatine for Use in Telomere Lengthening" at the U.S. Patent and Trademark Office (# 63/608,850). SMO has received research support related to creatine during the past 36 months from the Serbian Ministry of Education, Science, and Technological Development; Provincial Secretariat for Higher Education and Scientific Research; AlzChem GmbH; ThermoLife International; and Hueston Hennigan LLP. SMO is the co-founder of KRE-ALL (a company which develops food products enriched with creatine. No potential conflict of interest was reported by the author(s).

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Data availability statement

All data analyzed are included in the article. Further inquiries can be directed to the corresponding author.

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